

We want to map Energy to the Probability of bond breaking

for example: we saw : $\text{--}\dot{\text{C}}\text{--}\dot{\text{C}}\text{--}\dots\text{--}\dot{\text{C}}\text{--}\dot{\text{C}}\text{--}$ $\text{C}_{12}\text{H}_{26}$
but we didn't see: $\text{H--O--O}\dots\text{O--H}$ H_2O_N

Why? It has something to do with the Bond Dissociation Energy (BDE).

the BDE of O-O is 146 kJ/mol,

of C-C is 346 kJ/mol. C-C is 2.37 times STRONGER!

From this perspective, we can think of a mapping algorithm project BDE to $P_{\text{Bond Breaking}}$

ASSUMPTIONS

A few assumptions to be made:

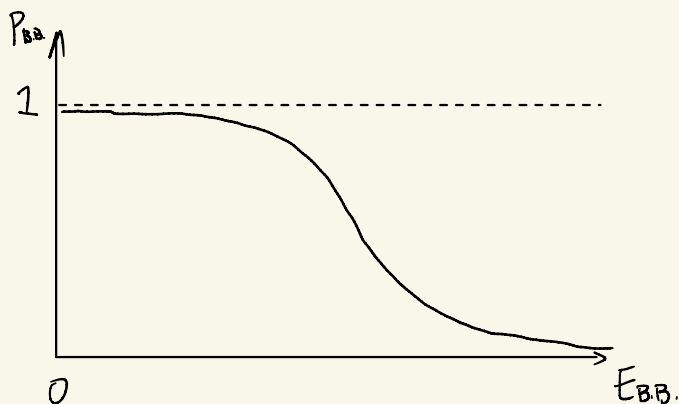
• neglect the difference between $\text{H}_3\text{C--CH}_3$ bond & $\text{H}_2\text{C--CH}_3$ bond

Therefore $\left\{ \begin{array}{l} \bullet \text{ We do not consider H-C, H-N, H-O breaking. Their strength are stronger than C-C, O-O, etc.} \end{array} \right.$

• We assume $P_{\text{molecule survive}} = \prod_i (1 - P_{\text{Bond Breaking}})$

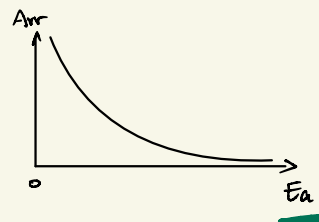
WHAT WE WANT

we want to map $E_{\text{BD}} \longrightarrow P_{\text{B.B.}}$

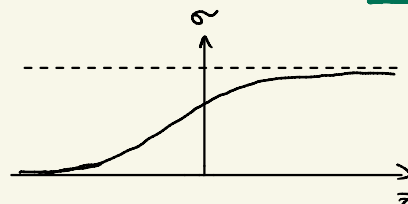


TOOLS

Arrhenius equation: $A_{\text{rr}} = A e^{-\frac{E_a}{RT}}$



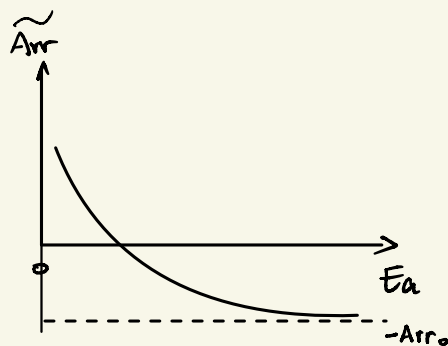
Sigmoid Function: $\sigma(z) = \frac{1}{1 + e^{-z}}$



ASSEMBLE TO WHAT WE WANT

① Introduce a correction term:

$$\tilde{A}_{rr} = A_{rr}(E_0) - A_{rr_0}$$

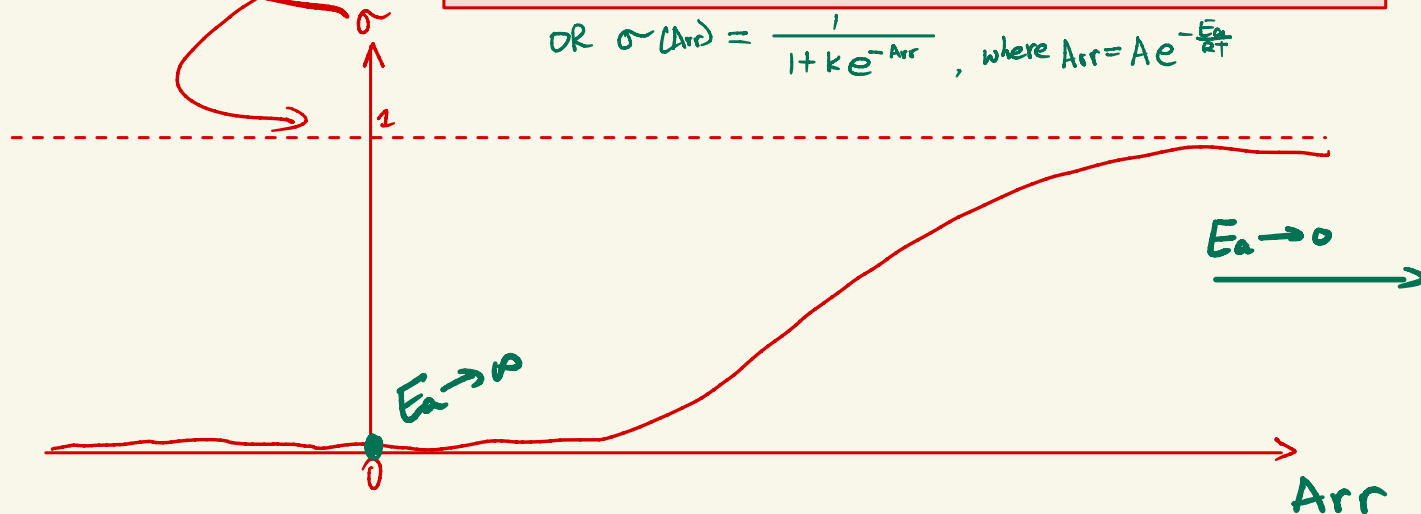


$$\textcircled{2} \quad \sigma(\tilde{A}_{rr}) = \frac{1}{1 + e^{-(Ae^{-\frac{E_a}{kT}} - A_{rr_0})}} = \frac{1}{1 + e^{-Ae^{-\frac{E_a}{kT}}} \cdot e^{A_{rr_0}}}$$

$$\sigma(A_{rr}(E_0)) = \frac{1}{1 + k e^{-Ae^{-\frac{E_a}{kT}}}}$$

OR $\sigma(A_{rr}) = \frac{1}{1 + k e^{-A_{rr}}}$, where $A_{rr} = Ae^{-\frac{E_a}{kT}}$

σ is the probability of Bond Breaking



for larger k , the curvature moves to the right.

Therefore, $P_{\text{single bond breaking}} = \sigma(A_{rr}(E_0))$

$$P_{\text{molecule survive}} = \prod (1 - P_{\text{s.B.B.}})$$